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GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

Compensators

GATE ECE Control Systems · Previous Year Questions with Answer Key

25	21	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	8 questions
Marks 2	14 questions
Marks 5	2 questions
Marks 8	1 question
Answer key	at the end

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Marks 1 8 questions

Q1. GATE ECE 2017 Set 2

Which of the following statement is incorrect?

(A) Lead compensator is used to reduce the settling time.

(B) Lag compensator is used to reduce the steady state error.

(C) Lead compensator may increase the order of a system.

(D) Lag compensator always stabilizes an unstable system.

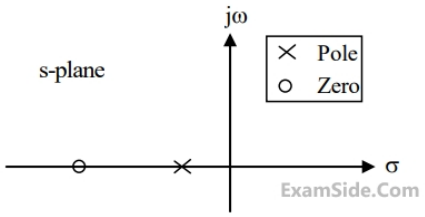
Open solution

Q2. GATE ECE 2017 Set 1

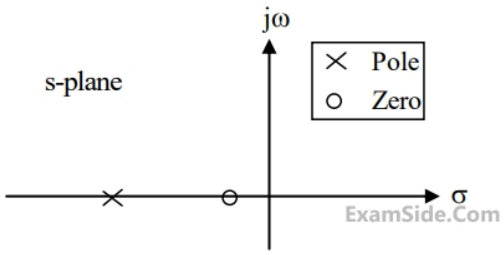
Which of the following can be the pole-zero configuration of a phase-lag controller (lag compensator)?

Open solution

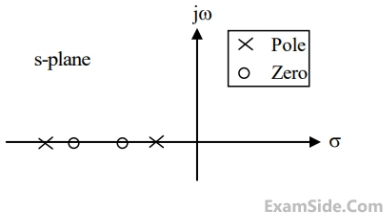
(A)



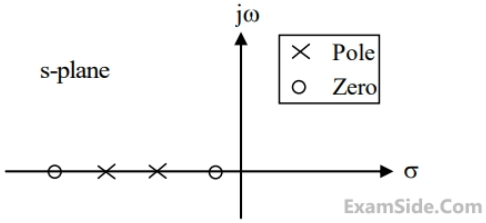
(B)



(C)



(D)



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Q3. GATE ECE 2015 Set 3

Open solution

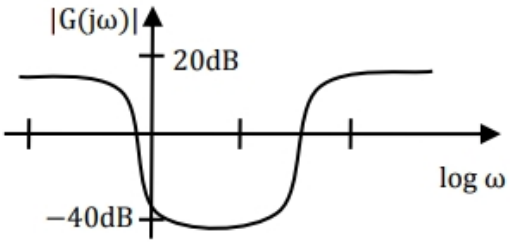
The transfer function of a first-order controller is given as $G_c(s) = \frac{K(s+a)}{s+b}$ where k, a and b are positive real numbers. The condition for this controller to act as a phase lead compensator is

(A) a < b (B) a > b (C) K < ab (D) K > ab

Q4. GATE ECE 2009

Open solution

The magnitude plot of a rational transfer function G(s), with real coefficient is shown in figure. Which of the following compensators has such a magnitude plot?



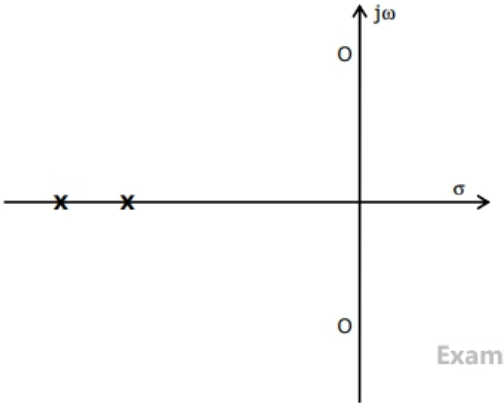
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(A) Lead compensator (B) Lag compensator
(C) PID controller (D) Lea-lag compensator

Q5. GATE ECE 2008

Open solution

The pole-zero plot given below corresponds to a



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(A) Low pass filter (B) High pass filter (C) Band pass filter (D) notch filter

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Q6. GATE ECE 2006

The transfer function of a phase-lead compensator is given by $G_c(s) = \frac{1+3Ts}{1+Ts}$ where T > 0. The maximum phase-shift provided by such a compesator is

Open solution

(A) $\pi/2$

(B) $\pi/3$

(C) $\pi/4$

(D) $\pi/6$

Q7. GATE ECE 2003

A PD controller is used to compensate a system. Compared to the uncompensated system, the compensated system has

Open solution

(A) a higher type number.

(B) reduced damping.

(C) higher noise amplification.

(D) larger transient overshoot.

Q8. GATE ECE 1998

The transfer function of a phase lead controller is $\frac{1+3Ts}{1+Ts}$. The maximum value of phase provided by this controller is

Open solution

(A) 90°

(B) 60°

(C) 45°

(D) 30°

Marks 2 14 questions

Q9. GATE ECE 2026

For the control system shown in the Figure, the transfer function of a plant, $G(s) = \frac{1}{(s+1)(s+2)}$ is connected in cascade with a compensator $C(s) = K(s+\alpha)$, where K and α are positive real valued constants.

Open solution

Which of the following pairs (K, α) represent the correct values for the closed loop system to have poles at $(-3 \pm j\sqrt{5})$?



(A) 2,3

(B) 3, 4

(C) 2,4

(D) 3,3

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Q10. GATE ECE 2025

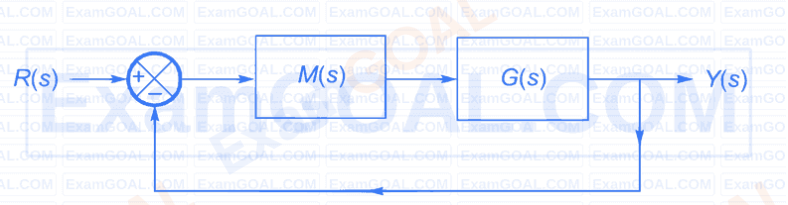
Open solution

Let $G(s) = \frac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to the system $G(s)$ in the configuration shown below. Consider the following statements.

(i) There exists no controller of the form $M(s) = \frac{K_I}{s}$, where K_I is a positive real number, such that the closed loop system is stable.

(ii) There exists at least one controller of the form $M(s) = K_P + sK_D$, where K_P and K_D are positive real numbers, such that the closed loop system is stable.

Which one of the following options is correct?



(A) (i) is TRUE and (ii) is FALSE

(B) (i) is FALSE and (ii) is TRUE

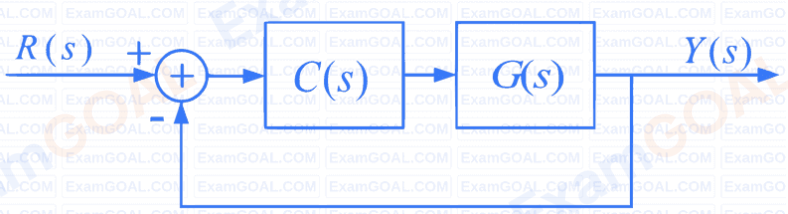
(C) Both (i) and (ii) are FALSE

(D) Both (i) and (ii) are TRUE

Q11. GATE ECE 2024

Open solution

A satellite attitude control system, as shown below, has a plant with transfer function $G(s) = \frac{1}{s^2}$ cascaded with a compensator $C(s) = \frac{K(s+\alpha)}{s+4}$, where K and α are positive real constants.



In order for the closed-loop system to have poles at $-1 \pm j\sqrt{3}$, the value of α must be _____.

(A) 0

(B) 1

(C) 2

(D) 3

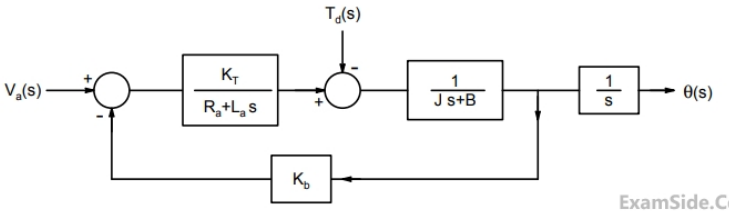
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Q12. GATE ECE 2015 Set 3

The position control of a **DC** servo-motor is given in the figure. The values of the parameters are K_t=1 N-m/A, R_a=1Ω, L_a=0.1H,

J=5kg-m^2, B=1 N-m/(rad/sec) and K_b=1V/(rad/sec).

The steady-state position response (in radians) due to unit impulse disturbance torque T_d is ____.



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Open solution

Q13. GATE ECE 2015 Set 1

A lead compensator network includes a parallel combination of 'R' and 'C' in the feed-forward path. If the transfer function of the compensator is $G_C(s) = \frac{s+2}{s+4}$. The value of RC is ____

Open solution

Q14. GATE ECE 2012

The transfer function of a compensator is given as $G_C(s) = \frac{s+a}{s+b}$.
$G_C(s)$ is a lead compensator if

(A) a = 1, b =2 (B) a = 3, b = 2 (C) a = -3, b = -1 (D) a = 3, b = 1

Open solution

Q15. GATE ECE 2012

The transfer function of a compensator is given as $G_C(s) = \frac{s+a}{s+b}$.
The phase of the above lead compensator is maximum at

(A) $\sqrt{2}$ rad/sec (B) $\sqrt{3}$ rad/sec (C) $\sqrt{6}$ rad/sec (D) $1/\sqrt{3}$ rad/sec

Open solution

Q16. GATE ECE 2010

A unity negative feedback closed loop system has a plant with the transfer function $G(s) = \frac{1}{s^2+2s+2}$ and a controller $G_c(s)$ in the feed forward path. For a unit set input, the transfer function of the controller that gives minimum steady sate error is

(A) $G_C(s) = \frac{s+1}{s+2}$

(B) $G_C(s) = \frac{s+2}{s+1}$

(C) $G_C(s) = \frac{(s+1)(s+4)}{(s+2)(s+3)}$

(D) $G_C(s) = 1 + \frac{2}{s} + 3_s$

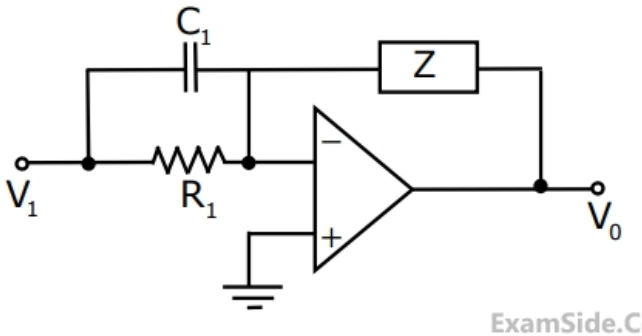
Open solution

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Q17. GATE ECE 2008

Group I gives two possible choices for the impedance Z in the diagram. The circuit elements in Z satisfy the condition $R_2C_2 > R_1C_1$. The transfer $\frac{V_0}{V_1}$ function represents a kind of controller. Match the impedances in **Group I** with the types of controllers in **Group II**.

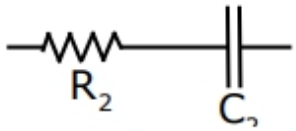
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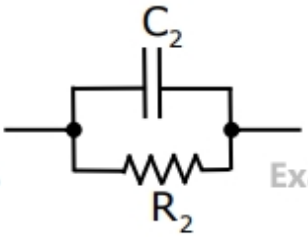
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Group - I

Q.



R.



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Group - II

1. PID controller
2. Lead compensator
3. Lag compensator

(A) Q - 1, R - 2 (B) Q - 1, R - 3 (C) Q - 2, R - 3 (D) Q - 3, R - 2

Q18. GATE ECE 2007

The open-loop transfer function of a plant is given as $G(s) = \frac{1}{s^2-1}$. If the plant is operated in a unity feedback configuration, then the lead compensator that can stabilize this control system is

Open solution

(A) $\frac{10(s-1)}{s+2}$ (B) $\frac{10(s+4)}{s+2}$

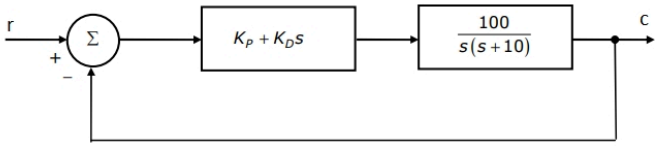
(C) $\frac{10(s+2)}{s+10}$ (D) $\frac{2(s+2)}{s+10}$

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Q19. GATE ECE 2007

A control system with a PD controller is shown in the figure. If the velocity error constant $K_v = 1000$ and the damping ratio $\zeta = 0.5$, then the values of K_P and K_D are

Open solution



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(A) $K_P = 100, K_D = 0.09$
(B) $K_P = 100, K_D = 0.9$
(C) $K_P = 10, K_D = 0.09$
(D) $K_P = 10, K_D = 0.9$

Q20. GATE ECE 2005

A double integrator plant, $G(s) = \frac{K}{s^2}, H(s) = 1$ is to be compensated to achieve the damping ratio $\zeta = 0.5$ and an undamped natural frequency, $\omega_n = 5$ rad/sec. Which one of the following compensator $G_c(s)$ will be suitable?

Open solution

(A) $\frac{s+3}{s+9.9}$
(B) $\frac{s+9.9}{s+3}$
(C) $\frac{s-6}{s+8.33}$
(D) $\frac{s+6}{s}$

Q21. GATE ECE 1992

A process with open-loop model $G(s) = \frac{Ke^{-s\tau_d}}{\tau s+1}$, is controlled by a PID controller. For this process

Open solution

(A) the integral mode improves transient performance
(B) the integral mode improves steady state performance
(C) the derivative mode improves transient performance
(D) the derivative mode improves steady state performance

Q22. GATE ECE 1990

The transfer function of a simple RC network functioning as a controller is: $G_c(s) = \frac{s+z_1}{s+p_1}$
The condition for the RC network to act as a phase lead controller is:

Open solution

(A) $p_1 < z_1$
(B) $p_1 = 0$
(C) $p_1 = z_1$
(D) $p_1 > z_1$

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Marks 5 2 questions

Q23. GATE ECE 2002

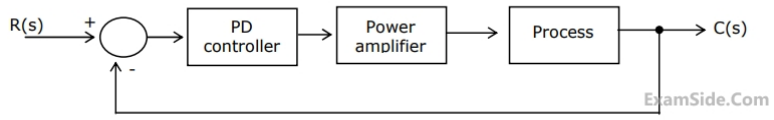
A unity feedback system has the plant transfer function $G_p(s)=\frac{1}{(s+1)(2s+1)}$

(a) Determine the frequency at which the plant has a phase lah of 90°.
(b) An intergral controller with transfer function $G_c(s)=\frac{k}{s}$, isplaced in the forwardpath the value of k such that the compensated system has an open loop gain margin of 2.5.
(c) Determine the steady state errors of the compensated system to unit-step and unit-ramp inputs.

Open solution

Q24. GATE ECE 2001

For the feedback control system shown in the figure, the process transfer function is $G_p(s) = \frac{1}{s(s+1)}$, and the amplification factor of the Power amplifier is $K \geq 0$. The design specifications required for the system, time constant is 1 sec and a damping ratio of 0.707.



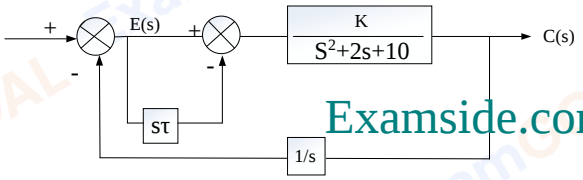
(a) Find the desired location of the closed loop poles.
(b) Write down the requiredcharacteristic equation for the system. Hense determine the PD controller transfer function $G_p(s)$ when K = 1.
(c) Sketch the root-locus for the system.

Open solution

Marks 8 1 questions

Q25. GATE ECE 1987

A system with derivative control is shown in figure.



(a)Determine the transfer function and the characteristic equation of the system.

(b)Given that K > 0, find the range of 'τ' for which the system is unconditionally stable.

Open solution

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Answer Key

Marks 1

1. **D**	2. **A**	3. **A**	4. **D**	5. **D**
6. **D**	7. **C**	8. **D**		

Marks 2

9. **B**	10. **D**	11. **B**	12. **-0.5**	13. **0.5**
14. **A**	15. **A**	16. **D**	17. **B**	18. **C**
19. **B**	20. **A**	21. **B, C**	22. **D**	

Marks 5

23.

(a) $\omega = \frac{1}{\sqrt{2}}$ **rad/sec**

(b) $K = \frac{7.5}{2}$

(c) Steady state error = $\frac{2}{7.5}$

24.

(a) The two poles are

w_1 = -1+1j.

w_2 = -1 -1j

(b) Characteristic equation, is

$s^2 + s\left(KK_D + 1\right) + KK_P = 0$

Marks 8

25.

(a) Characteristic equation is

$s^3 + 2s^2 + (10 + K\tau)s + K = 0$

(b) System is unconditionally stable when

$\tau > \frac{K-20}{2K}$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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